Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/06/2020
PI Name:	Pathak, Siddhartha Ph.D.		
Project Title:	Understanding the Local Structure-Property Relationships of Solders in Terrestrial vs. Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	50011	**Congressional District:**	4
Comments:	NOTE: PI moved in fall 2020 to Iowa State University from University of Nevada, Reno.		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
Start Date:	01/01/2020	**End Date:**	12/31/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Panda, Binayak	**Contact Phone:**	
Contact Email:	binayak.panda-1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: PI Dr. Sid Pathak moved to Iowa State University in fall 2020 and remains as Principal Investigator through a subcontract. Dr. Jeffrey LaCombe is now the administrative PI at University of Nevada Reno (Ed., 11/23/2020)		
Key Personnel Changes/Previous PI:	November 2020 report: The Principal Investigator (PI) S Pathak has moved from the University of Nevada, Reno (UNR) to Iowa State University (ISU), Ames. The primary PI at the UNR has been changed to Dr. Jeffrey LaCombe. UNR will sub-contract the funds for this project to Dr. Pathak at ISU.		
COI Name (Institution):	Asle Zaeem, Mohsen Ph.D. (Colorado School of Mines)		
Grant/Contract No.:	80NSSC20K0223		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The ongoing project is an integrated experimental and computational effort to characterize both the microstructure and resultant micro-to-nano mechanical response of solders in terrestrial vs. microgravity environments, 1 g vs. ~1×10^(-5) g. Results from the In-Space Soldering Investigation (ISSI) experiments performed aboard the International Space Station (ISS) have shown that soldering in microgravity is expected to be considerably different than their ground-based counterparts due to Earth's natural convective flow and buoyancy effects being minimized in microgravity during melting and solidification. The ISSI data has demonstrated that a lack of buoyancy forces in microgravity can internally trap the flux created during soldering at interfaces, such as repair joints. Our current hypothesis is that such internal porosity can be detrimental to the desired strength of the joint, as well as its thermal and electrical conductivity – our study aims to develop a comprehensive understanding of these effects in the millimeter-sized solders using an array of nano-mechanical testing tools such as indentation and focused ion beam (FIB) fabricated micro-pillar compression and micro tensile testing, and nanoscale electrical contact resistance (nano-ECR) measurements. Results from this study will be instrumental in enhancing our fundamental understanding of the effects of surface tension driven convection phenomena during solidification processing operations such as brazing, soldering, and welding. Furthermore, the microgravity experiments represent the lowest gravity boundary condition. As such, these results could also be useful in predicting solidification behavior on other lower gravity environments (e.g., moon or Mars). We have two graduate students working on a cutting-edge experimental-computational research project about soldering and solidification in microgravity conditions. One graduate student at Iowa State University, Ames is working on the experimental part, and another graduate student at Colorado School of Mines is working on the computational part of this project. We had two undergraduate senior design groups who worked on this project earlier.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ongoing project is unique in that it combines both experimental and computational research on solder samples from the micro-gravity and ground-based ISSI experiments, along with a comparative study on fresh solder samples. The insights gained from this work can be extended to study the effects of fluid interfaces with surface tension variations that are also common for many spaceflight-enabling technologies (e.g., boiling, heat transfer, combustion, welding, brazing, or soldering). We envision that this project will contribute to developing an advanced understanding of multiple processes that typically rely on gravity-driven separations in the terrestrial environment, such as the removal of gas bubbles from a liquid, which must be accomplished by other means in microgravity environments. Furthermore, the microgravity experiments represent a lowest gravity boundary condition. As such, these results could also be useful in predicting solidification behavior on other lower gravity environments (e.g., moon or Mars).
Task Progress:	The ongoing project is being conducted under two major thrusts: (A) microstructural characterization and (B) nanomechanical response of solders in terrestrial vs. microgravity environments. Each thrust is composed of integrated experimental (at University of Nevada, Reno UNR, and Iowa State University, ISU) and modeling (at Colorado School of Mines) tasks that leverage each other for further tuning and understanding. A breakdown of these two thrust areas is shown below: Thrust A.1. (UNR/ISU) Microstructure characterization of solders. Thrust A.2. (Mines) Develop an Advanced Phase-Field Model Capable of Predicting Solidification Defects/Pores and Mechanical Responses of Soldering in Microgravity and Terrestrial Conditions. Thrust B.1. (ISU) Small Scale Nano-mechanical Testing Strategies for Solders and Solder Joint Testing System (SJTS). Thrust B.2. Predicting mechanical response of solidified soldering specimens in microgravity and terrestrial conditions. Sufficient progress has been made in Year 1 on Thrusts A.1, A.2, and B.1. In year 2, work will continue on these thrusts, as well on Thrust B.2. Thrust A.1. It is essential to get an excellent and uniform surface to do the microstructural (EBSD analysis), and nanoindentation analysis as both the techniques are high surface-sensitive. Pb-Sn solder consists of two different Pb-rich and Sn-rich phases. Pb-Sn is a soft material, and pits forms during mechanical polishing because Pb has lower hardness as compared to the Sn phase. Scratches on the Pb-Sn surface generate very easily because of the softness of the solder. It is challenging to polish the solder sample because of its softness and two different phases in it. We could see scratches and pits in the SEM image even after 1 µm fine diamond polishing, so vibratory polishing (using 0.05 µm colloidal silica solution) is necessary as a final polishing step. After vibratory polishing, we could see the colloidal silica particle deposited on the sample surface. Normal cleaning using a cotton swab may generate scratches because of the softness of the sample material. Ultrasonic cleaning using distilled water is recommended for cleaning the Pb-Sn solder sample surface to avoid scratches during the cleaning process. We could not see any scratch or colloidal silica particle in the SEM image of the sample surface, but finer pits were visible. We have tried to get the polishing parameter to obtain the fine uniform surface, but further improvement in the polishing parameter is necessary. Microstructural and nano-mechanical testing is yet to be done. It has been delayed because of the lockdown imposed (From March 2020 to June 2020) due to the COVID pandemic and our group transfer from the University of Nevada, Reno to Iowa State University, Ames. The microstructure of the solders will be characterized using SEM techniques such as EBSD and EDS after polishing the samples to a very fine level. Nanoindentation testing will be done after microstructural characterization. We are planning to use both the sharp and spherical indenter tip geometries to correlate the local mechanical properties of the solders with the structure information obtained from microstructural characterization. Thrust A.2. The Present multiphase field model (MPF) model enables to predict the effects of alloying elements, solid-liquid interface property, and process parameters on phase formation, solidification patterns, and the possibility of micro-segregation of elements at each time step. A multiphase field model (MPF) simulation of the soldering process is expected to predict the effects of process parameters and microgravity conditions on defect and porosity evolutions. To this end, it is necessary to consider the fluid flow in micro-scale and interdendritic regions and buoyancy forces. In the next stage of developing our MPF model, the modified Navier-Stokes equations will be coupled to the present equations, enabling the model to predict the effect of gravity level on the micro-fluid flow in and void formation during the soldering process. Thrust B.1. The frame SJTS design has required a substantial amount of planning, thought, and reiteration. The result of this labor is a machine capable of testing surface mount solder joints for their shear strength in an accurate and repeatable way. The frame itself required CAD design to develop a professional and reliable machine suited to this

	testing. The individual frame parts were built from aluminum to improve the cost, machinability, and weight of the machine, while high-stress parts were built from steel. The testing tip and sample were carefully designed in a similar manner with the use of CAD design. By building this system so meticulously, the quality of the product was very high and predictable. The machining of the frame and tip were performed in-house to reduce cost and improve communication between the senior design team and the machinist. Although the focus for this type of experiment is on the data and analysis, it was shown how important the sample preparation was to receive consistent and accurate data during testing. Any failure to produce consistent samples could lead to differing failure mechanics and wide variances in the data. It was found that proper heating and wetting of the samples prior to soldering the components produced the most consistent samples. Care had to be taken, however, to prevent overheating and burning the components such that they failed to operate. A visual inspection of every sample was completed to ensure consistency prior to testing. The data from testing was acquired through the Arduino, which organized and output the data values. The Realterm software was then used to convert the output data into a .csv format, which allowed for easier data analysis when imported into Excel. Finally, Excel allowed for a visual representation of the data into load vs. displacement graphs. It was discovered that the load vs. displacement graphs were not providing useful information due to limitations within the actuator hardware. It is possible that a workaround could be found within the Arduino software, and that may be something that a future senior design group may be able to resolve. The load vs. time graphs were sufficient to prove that the solder joint testing system (SJTS) was successfully designed to measure the loads required to cause failure in the surface mount solder joints. Unfortunately, the lack of resources available during testing meant that the sample size of solder joints was smaller than planned. However, the load data shows that an average force of 49 N for the smaller 0603 resistors and 66 N for the larger 1206 resistors was in line with the values that were expected. The plan for the analysis of the solder joints included visual inspection with an optical microscope. This analysis would have allowed for a greater understanding of the types of shear failure modes that occurred during testing. This was not possible due to a lack of access to equipment on the university campus. Should the project continue, and samples produced in reduced gravity are tested, it will be necessary to complete this analysis and extend it to the reduced gravity samples. The failure mode of the solder joint is a critical piece of information for understanding the way reduced gravity affects these solder joints. Overall, the SJTS has proven that, given the proper resources, it is a capable and reliable machine for testing the mechanical properties and reliability of solder joint samples. It is expected that this will apply to future samples testing using this system that are produced in reduced gravity aboard the International Space Station. Thrust B.2. Planned for year 2.
Bibliography Type:	Description: (Last Updated: 04/22/2025)
Dissertations and Theses	Dietrich R, Gutierrez J, Michel M. "Senior Design Project 2020: Solder Joint Testing System." A design report presented in partial fulfillment for the degree of Bachelor of Science in Materials Science and Engineering, University of Nevada Reno, June 2020 , Jun-2020